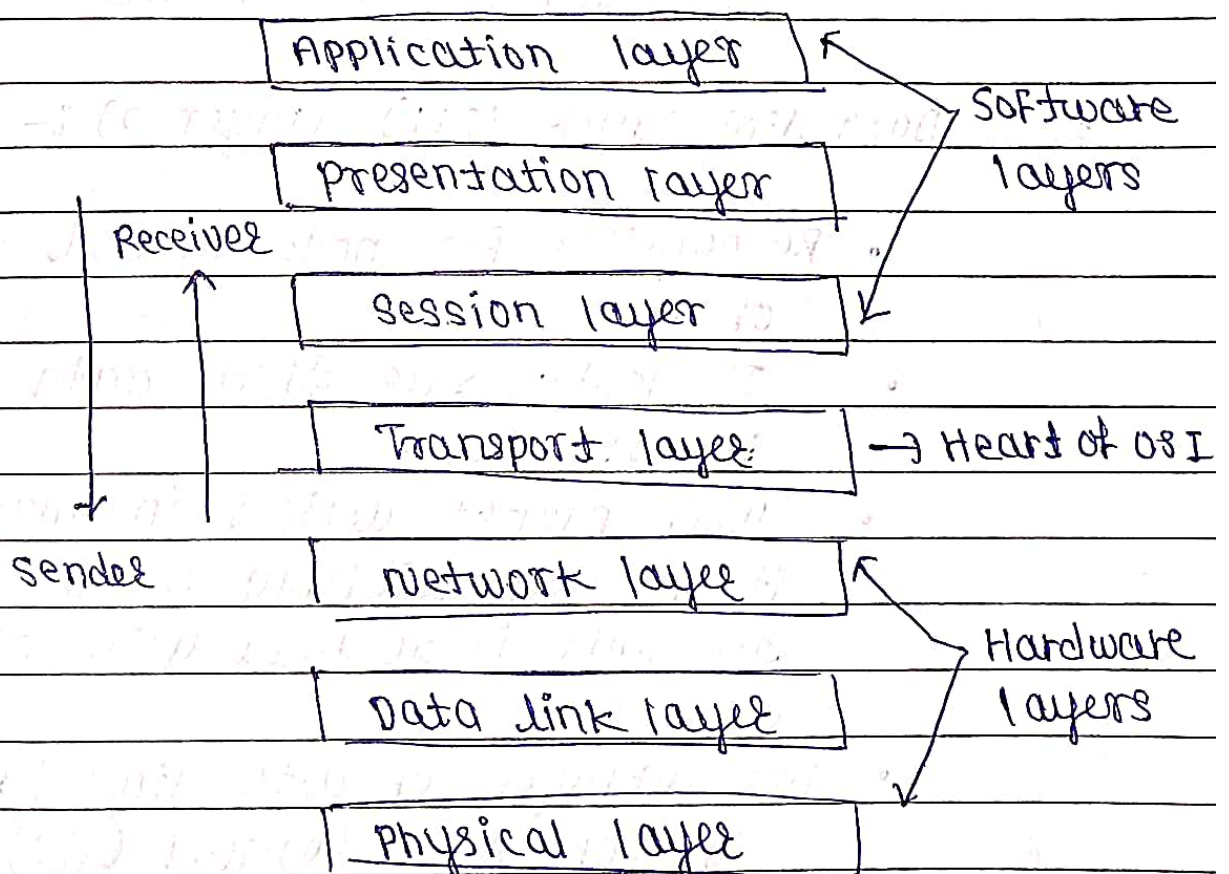


UNIT V :- OSI REFERENCE MODEL & TCP/IP MODEL

5.1] Layers of OSI Reference Model :-

- OSI - Open System Interconnection
- Developed by - International organization for standardization in 1984
- It is 7 layer architecture with each layer having specific functionality to perform.
- All these layers work collaboratively to transmit data from one person to another across globe.



1] Physical layer (layer 1) :-

- lowest layer of OSI model
- Responsible for actual physical connection between devices
- It contains information in form of bits & transmits bits from one node to next.
- It receives signal, converts it into 0s & 1s & sends to data link layer.
- Functions :-
 - i) Bit synchronization
 - ii) Bit rate control
 - iii) Physical topologies
 - iv) Transmission mode

2] Data link layer (DLL) (layer 2) :-

- Responsible for node to node delivery of msg
- It makes sure that data transfer is error free.
- When packet arrives in network, it is responsibility of DLL to transmit it to host using ~~MAC~~ MAC address.
- Two sublayers of data link layer:-
 - i) Logic link control (LLC)
 - ii) Media Access control (MAC)
- Functions :-
 - i) Framing

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- ii) physical addressing
 - iii) Error control
 - iv) Flow control
 - v) Access control

3) network layer (layer 3) :-

- works ~~for~~ for transmission of data from one host to other located in different networks
- It also take care of packet routing
- sender & Receiver's IP addresses are placed in header by network layer.
- Functions :-
 - i) Routing
 - ii) logical addressing

4) Transport layer (layer 4) :-

- Provides services to Application layer & takes services from network layer.
- Data in transport layer known as segments
- It is responsible for end to end delivery of complete msg.
- It retransmits data if error is found.
- At receiver's side - TL provides services to application layer & takes services from n/w layer
- At ~~re~~ sender's side :- TL receives msg from upper layers into packets &

resembles these packets again into msg at destination.

- Provide two types of services:-

1) connection oriented transmission:-

Sends acknowledgement to source

2) connectionless transmission:-

doesn't send acknowledgement

- Functions:-

i) Segmentation & Reassembly

ii) Service point Addressing

5] Session layer (layer 5):-

- Responsible for establishment of connection,
- maintenance of sessions, authentication
- ensures security

- Functions:-

i) session establishment, maintenance, & termination

ii) Synchronization

iii) Dialog controller

6] presentation layer :- (layer 6)

- Also known as Translation Layer
- Data from application layer is extracted here & manipulated as per the required format to transmit over the network.

- Functions:-

i) Translation

ii) Encryption / Decryption

iii) Compression

7] Application layer (layer 7) :-

- Top layer of OSI
- produce data which has to be transfer over network
- Acts as window for application services to access the network & for displaying the received information to the user.
- ex - Browser, skype, messenger etc.
- Functions -
 - i) network virtual Terminal
 - ii) FTAM - File Transfer & management
 - iii) Mail services
 - iv) Directory Services
- OSI model is a reference model & is not implemented bcoz of it's late invention. The current model being used is TCP/IP model.

5.2] LAYERED STRUCTURE OF TCP/IP MODEL :-

- TCP/IP :- Transmission control protocol / Internet protocol
- Developed by - Department of Defense (DoD) in 1960
- OSI is just reference model just to describe function of communication system
- TCP/IP is concise version of OSI model with 4 layers -

1] Application layer (4th layer) ^{Network Access layer}

- combination of physical layer & data link layer of OSI model
- It looks out for hardware addressing & protocols present in this layer allows for physical transmission of data.

2] Internet layer :-

- parallels function of OSI's network layer
- defines protocol which are responsible for logical transmission of data over entire network.
- main protocols residing in this layer are :-

- 1) IP - Internet protocol
- 2) ICMP - Internet control msg protocol
- 3) ARP - Address Resolution Protocol

3] HOST TO HOST LAYER :-

- Analogous to transport layer of OSI
- Responsible for end to end communication & error free delivery
- It shields upper layer applications from complexities of data
- Two main protocols in this layer are :-
 - 1) TCP - Transmission control protocol known to provide reliable & error free communication between end systems
 - 2) UDP - User datagram protocol: Connectionless protocol

4] Application layer :-

- Performs functions of top 3 layers of OSI model
- Responsible for node to node communication
- Controls user interface specifications
- Some of protocols present in this layer are - HTTP, HTTPS, NTP, DNS, NFS, SSH, Telnet, SMTP, & DHCP
- 1) HTTP :- Hyper Text Transfer protocol Used by world wide web to manage communications between web browsers & servers

2] SSH :- secure shell

It sets up secure session over TCP/IP connection.

3] NTP :- network time protocol
used to synchronize clocks on
our computer

5.3]

TCP / IP

OSI

i] Transmission control
protocol

Open system
Interconnection

ii] 4 layers

7 layers

iii] more reliable

Less reliable

iv] doesn't have very
strict boundaries

Have strict
boundaries

v] Follow horizontal
approach

Follow vertical
approach

vi] Uses both session
& presentation layer
in application
layer

uses different
session and
presentation layers

vii] Developed protocols
then
model

Developed model
then protocol

viii]	Transport layer doesn't provide assurance delivery of packets	Transport layer provides assurance delivery of packets
ix]	network layer only provides connectionless services	network layer provides both connection less and connection oriented services
x]	Protocols can't be replaced easily	Protocols can replace easily.

5.4] IP Addressing :-

* IPv4 address :-

- 32 bit long
- Unique & universal
- address space is 2^{32} ~~or 4,294,967,296~~

* change the Following IPv4 addresses From binary notation to dotted decimal notation

a) 10000001 00001011 00001011 11101111
 → 128+1 8+2+1 8+2+1 128+64+32+8+4+2+1
 129 11 11 239

2⁷ 2⁶ 2⁵ 2⁴ 2³ 2² 2¹ 2⁰
 128 64 32 16 8 4 2 1

129.11.11.239

* Change following IPv4 address From dotted-decimal notation to binary notation

a) 111.56.45.78

Q	R	Q	R	Q	R	Q	R
111/2	55 1	56/2	28 0	45/2	22 1	78/2	39 0
55/2	27 1	28/2	14 0	22/2	11 0	39/2	19 1
27/2	13 1	14/2	7 0	11/2	5 1	19/2	9 1
13/2	6 1	7/2	3 1	5/2	2 1	9/2	4 1
6/2	3 0	3/2	1 1	2/2	1 0	4/2	2 0
3/2	1 1	1/2	0 1	1/2	0 1	2/2	1 0
1/2	0 1					1/2	0 1

1101111.111000.101101.1001110
 01101111.00111000.00101101.01001110

Q Find error in following IPV4 addresses.

a) 111.56.045.78

→ There must be no leading zero (045)

b) 221.34.7.8.20

→ There can be no more than four no's

c) 75.45.301.14

→ Each number needs to be less than ^{or equal to} 255

d) 11100010.23.14.67

→ mixture of binary & decimal no's is not allowed

• Finding classes in binary & dotted decimal notation.

class	Binary	decimal
A	0	0-127
B	10	128-191
C	110	192-223
D	111 0	224-239
E	1111	240-255

Q Find class of each address -

a) 000 00001 00001011 00001011 1110 1111
→ class A (First bit is 0)

b) 110 00001 10000011 00011011 11111111
→ ~~class~~ class C (First 2 bytes 1 & third byte is 0)

c) 14.23.120.8
→ class A First byte is 14

d) 252.5.15.111
→ class E (First byte is 252)

• Default mask for classful addressing

class	Binary	Dotted decimal	CIDR
A	11111111 00000000 00000000 00000000	255.0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255.0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255.0	/24

- If ~~the~~ block of 16 addresses ~~is~~ is granted to small organization,

- First address can be calculated by setting rightmost 32-n bits to 0s
- Last address can be calculated by setting rightmost 32-n bits to 1s
- general form - x.y.z.t/n (n defines mask)

Q] Block of 16 addresses is granted to small organization. One of addresses is 205.16.37.39/28
Find first & last address of this block

205.16.37.39/28

↓ Binary

11001101 00010000 00100101 00100111

1st address:-

11001101 00010000 00100101 00100000

or 205.16.37.32

Last address:-

11001101 00010000 00100101 00101111

or 205.16.37.47

no. of address in blocks =

$$2^{32-n}$$

$$= 2^{32-28}$$

$$= 16 \text{ addresses.}$$

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* Another way to find address

In example 5, 128 can be write as-

1111111 1111111 1111111 1110000
(28 ^{1s} & Four 0s)

Find First address, last address & no. of addresses

→ 1] First address :-

AND address with mask

Address : 11001101 00010000 00100101 00100111

Mask : 11111111 11111111 11111111 11110000

1st address: 11001101 00010000 00100101 00100000

2] Last address :-

ORing address with complement of mask

Address : 11001101 00010000 00100101 00100111

Mask : 00000000 00000000 00000000 00001111

Last address: 11001101 00010000 00100101 00101111

3] no. of addresses :-

complement mask, interpret it as decimal no & add 1 to it

mask complement: 00000000 00000000 00000000

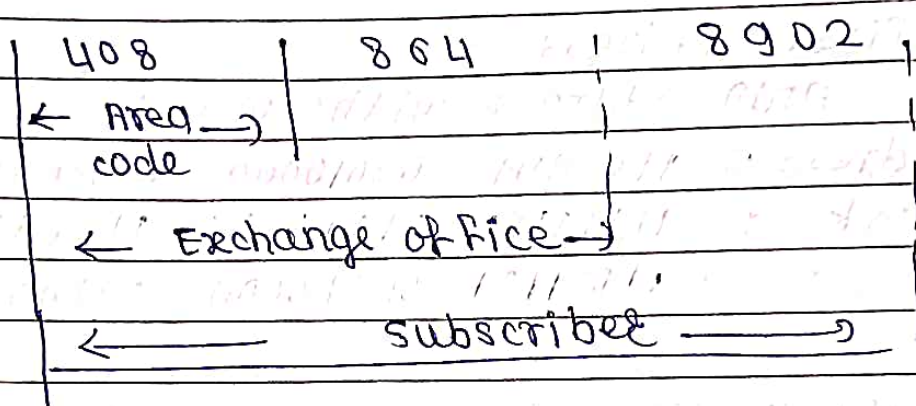
00001111

no. of addresses : $15+1=16$

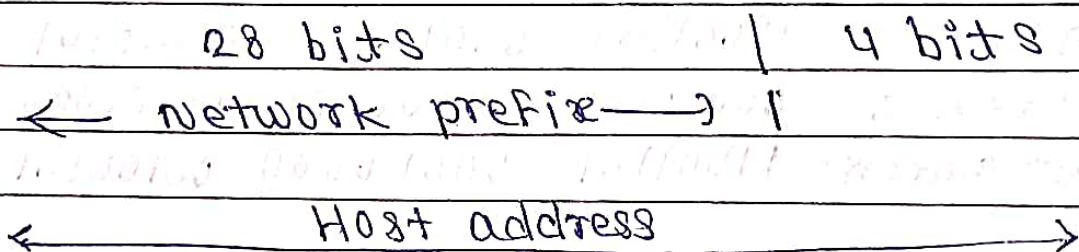
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Note:-

- The First address in a block is normally not assigned to any device, it is used as network address that represents the organization to the rest of the world.
- Two levels of hierarchy in IPv4 address:-



- A Frame in character oriented protocol:-



Note:-

each address in block can be considered as two-level hierarchical structure:

the leftmost n bits (prefix) define the network;

the rightmost $32-n$ bits define the host

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* Example :-

An ISP granted block of addresses starting with $190.100.0.0/16$ (63,536 addresses). ISP wants to distribute these to three grp of customers as follows:-

a] First grp:- 64 customers
each needs 256 addresses

this means 8 ($\log_2 256$) bits are needed to define each host.

$$\text{prefix length} = 32 - 8 = 24$$

$\therefore$ addresses are

1st customer: $190.100.0.0/24$ $190.100.0.255/24$

2nd customer: $190.100.1.0/24$ $190.100.1.255/24$

...

64th customer: $190.100.63.0/24$ $190.100.63.255/24$

$$\text{Total} = 64 \times 256 = 16,384$$

b] 2nd grp: 128 customers
each needs 128 addresses

this means 7 ($\log_2 128$) bits are needed to define each host

$$\text{prefix length} = 32 - 7 = 25$$

$\therefore$ addresses are-

1st customer: 190.100.64.0/25 190.100.64.127/25
 2nd customer: 190.100.64.128/25 190.100.64.255/25

 128th customer: 190.100.127.128/25 190.100.127.255/25
 Total = $128 \times 128 = 16,384$

c) 3rd grp :- 128 customers
 each needs 64 addresses

This means that 6 ($\log_2 64$) bits
 are needed to each host.

prefix length = $32 - 6 = 26$

$\therefore$ addresses are:-

1st customer: 190.100.128.0/26 190.100.128.63/26
 2nd customer: 190.100.128.64/26 190.100.128.127/26

 128th customer: 190.100.159.192/26 190.100.159.255/26
 Total = $128 \times 64 = 8192$

no. of granted addresses to ISP : 63,536

no. of allocated address by ISP : 40,960

available address :- 24,576

* Five column translation table:

Private Address	private port	External Address	External port	Transport protocol
172.18.3.1	1400	25.8.3.2	80	TCP
172.18.3.2	1401	25.8.3.2	80	TCP

* IPV6 Address :-

• 128 bits long

128 bits = 16 bytes = 32 hex digits

original

FDEC : 0074 : 0000 : 0000 : 0000 : BOFF : 0000 : FFF0

Abbreviated

FDEC : 74 : 0 : 0 : 0 : BOFF : 0 : FFF0

more abbreviated

FDEC : 74 :: BOFF : 0 : FFF0

↑ gap

Ex -

Expand address to its original -

0:15::1:12:1213

0000:0015:0000:0000:0000:0001:0012:1213

First align left side of double colon to left original pattern then align right side.

* IP address :-

- unique address that identifies device on the internet or local network
- IP stands for internet protocol → which is set of rules governing the format of data sent via internet or local network.
- Computer need address to communicate
- It is string of no's
- Full IP address ranges from 0.0.0.0 to 255.255.255.255
- IP address are assigned by IANA (Internet corporation For Internet Assigned no's authority)
- IP address changes according as you change network of your device.
- Two types of IP addresses:
IPv4 & IPv6

IPV4

IPV6

i]	32 bit address length	128 bit address length
ii]	supports manual & DHCP address configuration	supports Auto & renumber configuration
iii]	End to end connection integrity is unachievable	End to end, connection achievable
iv]	It can generate 4.29×10^9 address space	3.4×10^{38} address space
v]	Security Feature is dependent on application address	IPSEC is an inbuilt Security Feature
vi]	Address representation is decimal	— It — hexadecimal
vii]	Fragmentation performed by sender & forwarding routers	— It — by sender
viii]	Packet Flow identification is not available	— It — is available

ix]	checksum Field is available	— 11 — not available
x]	Broadcast msg transmission scheme	multicast & anycast transmission scheme
xi]	Encryption & authentication facility not provided	— 11 — provided
xii]	Header of 20-60 bytes	header of 40 bytes fixed
xiii]	IPv4 can be converted to IPv6	Not all IPv6 are converted to IPv4
xiv]	consists of 4 Fields separated by dots (.)	consists of 8 Fields separated by colons (:))
xv]	Divided into 5 different classes- A, B, C, D & E	does not have any classes
xvi]	Supports VLSM (variable length subnet mask)	does not support VLSM
xvii]	Ex- 66.94.29.13	Ex- 2001: 0000: 3238: DFE1 :00 0000: 0000: 22F1

* IP address classes:

A B C D E

* ~~classless IP addressing:-~~

* Network address:

Identifies network on internet

* Mask:

32 bit binary no that gives network address in address block

* Subnetting:-

Dividing large blocks of addresses into several contiguous sub-blocks & assigning these sub-blocks to different smaller networks is called subnetting. It is practice that is widely used when classless addressing is done.

* classless Addressing:-

Addressing architecture that uses variable length subnet masking.

* supernetting:-

procedure of combining multiple subnetworks into one network.

5.5]

1) DHCP :-

- Dynamic Host configuration protocol
- It is application layer protocol which is used to provide:

Subnet mask, Router address, DNS address, vendor class Identifier.

- DHCP based on client-server network model.
- DHCP port no. for server is 67 & for client is 68.
- It is client server protocol which uses UDP services.
- In DHCP, client & server exchange mainly 4 DHCP msgs in order to make connection, also called DORA process, but there are 8 DHCP msgs in the process.

• Advantages:

- i) Centralized management of IP addresses
- ii) Ease of adding new clients to a network
- iii) simple reconfiguration of IP addresses

• Disadvantages:-

- i) IP conflict can occur

2] NAT :-

- Network Address Translation
- To access internet, one public IP address is needed but we can make use a private IP address in our private network.
- Idea of NAT allow multiple devices to access internet through single public address.
- To achieve this, translation of private IP address to public IP address is required
- NAT is process in which one or more local IP address is translated into one or more Global IP address & vice versa in order to provide Internet access to local hosts.
- It also does translation of port numbers.
- It then makes corresponding entries of IP address & port number in NAT table
- NAT generally operates as router or firewall.

• Advantages -

- i) conserves legally registered IP addresses
- ii) provides privacy
- iii) Eliminates address renumbering when network evolves.

- Disadvantages :-

- i) Translation results in switching path delays
- ii) certain applications will not function on enabling NAT
- iii) Complicate tunneling protocols such as - IPsec